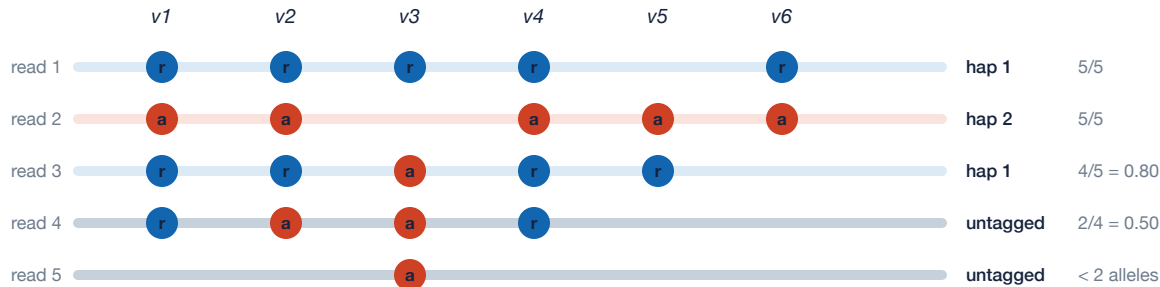


a Step 1 — assign each read to a haplotype



tagged if $\max(n_1, n_2) / (n_1 + n_2) > 0.65$ and weighted evidence > 1

allele weights: SNV 1, SV 1, indel 0.1, 5mC 0 (so ≥ 2 SNV/SV alleles are needed)

b Step 2 — re-derive each variant's phase from tagged reads

variant v₃

$n_1 = 2, n_2 = 1$

n_1 : hap-1 REF + hap-2 ALT reads

n_2 : hap-1 ALT + hap-2 REF reads

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2} = \frac{2}{3} = 0.67$$

$\leq 0.75 \rightarrow$ unphased, GT 0/1

variant v₁

$n_1 = 3, n_2 = 0$

n_1 : hap-1 REF + hap-2 ALT reads

n_2 : hap-1 ALT + hap-2 REF reads

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2} = \frac{3}{3} = 1.00$$

$> 0.75 \rightarrow$ phased, GT 0|1